

Problem ID: spiralstaircases

You started your new project: Installing a home automation system. You already bought all the components and in your local electronic shop you found a promotion set with a bunch of cables with different lengths.

Now you have to connect your controller with your smart sandwich maker that is several meters away and you are lacking a cable that is long enough.

To solve this problem you have to connect some of your wires to form a long one. You measure the lengths of every wire you own. The isolation is stripped on both ends of every wire for exactly 5 centimeters. To connect two wires, you overlap and twist the stripped ends. It is enough for the wires to touch each other with an overlap of 0. You cannot have an overlap of more than 5 centimeters, but the connection quality increases with longer overlaps. On both ends — the controller and the sandwich maker — you have a 5 centimeter stripped end as well to which you have to connect your newly created wire in the same way.

The connection quality of your link is determined by the smallest overlap used and your goal is to maximize this value.

The problem would be really easy, but your perfectionist wife hates unnecessary use of wires. Therefore the wire has to form a straight line without any loops and detours. And cutting a wire is no option, obviously.



Figure 1: 4 wires of different lengths connect the controller with the sandwich maker with different overlappings. Connection 1 is the maximal overlap, connection 2 the minimal overlap and all other connections are in between. The quality of this setup is 0.

Considering all possible arrangements of wires, find the one with the best quality.

Input

The first line of input consists of two integers n and d , the number of wires available and the distance to cover. This distance is measured between the casing of the controller and the sandwich maker. Each of the next n lines contain an integer d_i denoting the length of wire i including the stripped ends.

- $1 \leq n \leq 60$
- $11 \leq d \leq 1000$
- $11 \leq d_i \leq 1000$

Output

Output the best achievable quality. The quality must be accurate up to a relative or absolute error (whichever is lower) of 10^{-7} . If no arrangement fits your needs, output one line containing "impossible".

Sample Input 1

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3 70
20
35
50
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Sample Output 1

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3.3333333
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